THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

NMTC at SUB JUNIOR LEVEL - VII & VIII Standards

Saturday, 12 February August, 2022

Time : 2 Hrs. **M. M. : 30**

Note:

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.

2. Diagrams are only visual aids; they are NOT drawn to scale.

3. You are free to do rough work on separate sheets.

4. Duration of the test: **2 hours.**

PART—A

Note

- Only one of the choices A. B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering; seek the guidance of the supervisor.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**

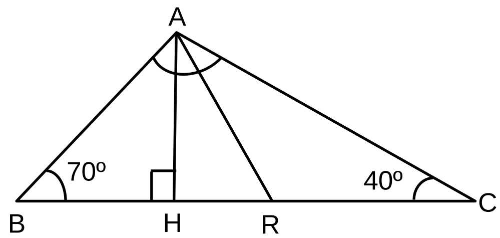
1. If x + y = 6, the value of $(x^3 + y^3 + 18xy)$ is

(a) 94 (b) 100 (c) 216 (d) 200

2. The radius and vertical height of a cone are respectively r and 2h. The radius and height of a cylinder are respectively 2r and h. If V_1 and V_2 are respectively the volumes of the cone and the cylinder, then $V_1 : V_2$ is.

(a) 1 : 3 (b) 1 : 4 (c) 1 : 6 (d) 1 : 5

3. In the adjoining figure AR is the bisector of $\angle$BAC and AH is the altitude. The measure of $\angle$HAR is



(a) 20º (b) 15º (c) 28º (d) 10º

4. The value of x satisfying the equation $\frac{3x-4}{2x+1} = \frac{6x-3}{4x+7}$ is

(a) a multiple of 3 (b) a multiple of 7 (c) a multiple of 4 (d) a prime number

5. Given n = 2021, the value of $\frac{7^n + 7^{n+1}}{7^{n+2} - 7^n}$ is

(a) $\frac{1}{6}$ (b) $\frac{1}{8}$ (c) $\frac{2}{9}$ (d) $\frac{1}{7}$

6. If a + b + c = 0, the numerical value of $\frac{a^2}{2a^2+bc}+\frac{b^2}{2b^2+ca}+\frac{c^2}{2c^2+ab}$ is

(a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$

7. A cuboid has length a units, breadth b units and height c units. S, V are the total surface area and volume of the cuboid respectively.

Then the numerical value of $\frac{V}{S}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ is

(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$

8. Samrudh divided $\frac{7}{9}$ by $\frac{2}{3}$ and got the answer x ; Saket divided $\frac{5}{11}$ by $\frac{15}{22}$ and got the answer y; Vishwa divided x by y ; the answer he got is

(a) $\frac{7}{4}$ (b) $\frac{6}{7}$ (c) $\frac{2}{9}$ (d) $\frac{1}{9}$

9. The simplified value of $\sqrt{\frac{(2022)^3-(2020)^3-2}{6}}$ is

(a) 2021 (b) 1 (c) 3 (d) $(2021)^2$

10. If X^{x^5} = 5, then the value of x is

(a) $5^{1/25}$ (b) 5^5 (c) $\left(\frac{1}{5}\right)^5$ (d) $5^{1/5}$

11. If a ≠ b ≠ c, $a^2 + a = b^2$, $b^2 + b = c^2$ and $c^2 + c = a^2$, then the numerical value of (a – b) (b – c) (c – a) is

(a) 1 (b) 0 (c) 2 (d) abc

12. The radius of the circle inscribed in a regular hexagon of side 10 units is r ; then

(a) 6 < r < 7 (b) 7 < r < 8 (c) 8 < r < 9 (d) 9 < r < 10

13. Each digit of a large number P is 1. If the number of digits of P is 2021, the value of the sum of the digits of the product P × 2021 is

(a) 2021 (b) 2021^2 (c) 202120 (d) 10105

14. A retail merchant goes to a wholesale supplier of goods with a certain sum of money, with which he can buy either 100 pens or 80 pencils. Retaining with himself 20% of money for conveyance to go back home, he buys 40 pencils and of the balance he purchases pens. The number of pens he bought is

(a) 40 (b) 50 (c) 30 (d) 35

15. If $\frac{2}{n}+\frac{1}{n^2}$ (where n is a natural number) can be written as an irreducible fraction whose numerator and denominator has no common factor and the sum of the numerator and denominator is 2025; then the numerator is

(a) 63 (b) 64 (c) 91 (d) 89

PART - B

Note :

- Write the correct answer in the space provided in the response sheet
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ marks.

16. There are baskets of 500 mangoes sold by 4 roadside vendors. Four persons A, B, C and D bought mangoes, each from a different vendor. A got 10% less than the number of mangoes bought by B; B got 25% more than the number of mangoes got by C while C got 20% less than D's number of mangoes. If A got 360 mangoes, then D got x% of mangoes in basket. Then the value of x is _______.

17. The ratio of an interior angle of the exterior angle of a regular polygon is 5 : 1. The number of sides of the polygon is ________.

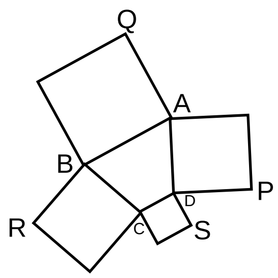
18. Some cows and some hens, totally 20 in all, were seen in a place. When all the legs are counted we get 64. The number of cows is _______.

19. Given a, b, c, d, e are natural numbers such that a + b = b + c = c + d = d + e = 4021 and a + b + c + d + e = 11063
Then the numerical value of (a – d) is _______.

20. The sum of the digits of a two-digit number is 6. If we add 18 to this number, we get a two-digit number consisting of the same digits in the reverse order. The sum of the squares of the digits of the original two-digit number is_____.

21. The four natural numbers n, m, n-m, n + m (where n > m) are all primes. The least numerical value of the sum of all these primes is ______.

22. In the adjoining figure, ABCD is a quadrilateral. Squares are drawn to each side as in the figure.



AP = $4\sqrt{2}$ units
Area of ΔABQ = 18 (\units)^2.
Area of square on BC is 25 (units)^2 and the perimeter of the square on CD is 18 units.

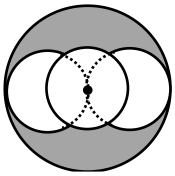
The perimeter of the quadrilateral ABCD (in the same units) is $\frac{p}{q}$ where p, q are natural numbers which have no common then the value of (p + q) is _______.

23. My friend Mr. Jack, has reached a square age. Mrs. Jack's age is the product of the digits of her husband's age. They have a son and a daughter. The age of the son is the sum of the digits of the age of the mother while the age of the daughter is the sum of the father. The age of the son is ______.

24. The lengths of the sides of a triangle (in centimeters) are consecutive integers. The length of the shortest side is 25% of its perimeter. The area of the triangle in square centimeter is ______.

25. Amar lists some four positive integers in some order, such that 1 more than the first, 2 less than the second, 3 times the third and one-fourth of the last number are all the same. If the sum of the four numbers is 58, the largest number in the list is_______.

26. In the adjoining figure, the radius of the largest circle is 2 cm and the radius of each of the three equal circles is 1 cm. The middle most circle is concentric with the largest one. the area of the shaded portion is $\frac{7\pi - 3\sqrt{a}}{3}$ square units. Then the value of 'a' is _____



27. There are four two digit numbers. None of the digits is zero ; no two of them are the same. In all the 8 digits used, one digit is not included. If their sum is 221, the excluded digit is _______.

28. If a, b, c are natural numbers such that s $a + \frac{1}{b + \frac{1}{c}} = \frac{37}{16}$, then the numerical value of a + b + c is _______.

29. Given a, b, c are real numbers such that $a - b = 8$ and $ab + c^2 + 16 = 0$.
The numerical value of $a^{2021} + b^{2021} + c^{2021}$ is

30. The value of $\left(\frac{a}{a^2 - 4} - \frac{8}{a^2 + 2a}\right)\left(\frac{a^2 - 2a}{4 - a}\right) + \frac{a + 8}{a + 2}$ is given to be 1.
The value of 'a' is _______.